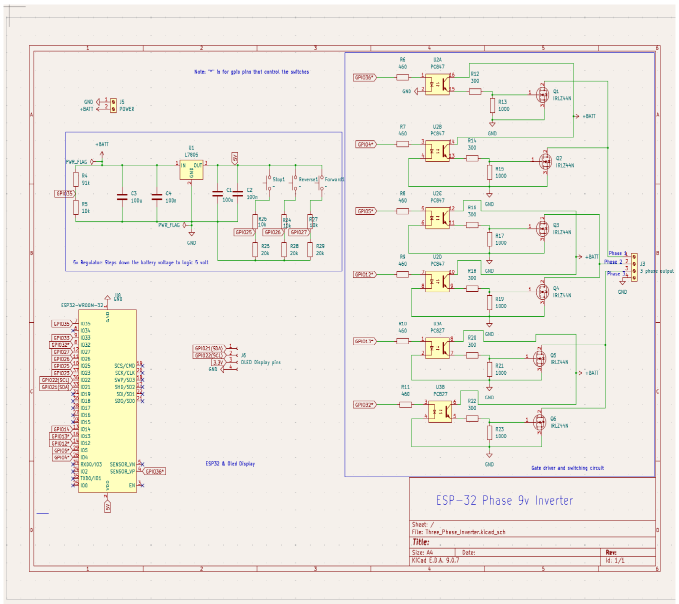
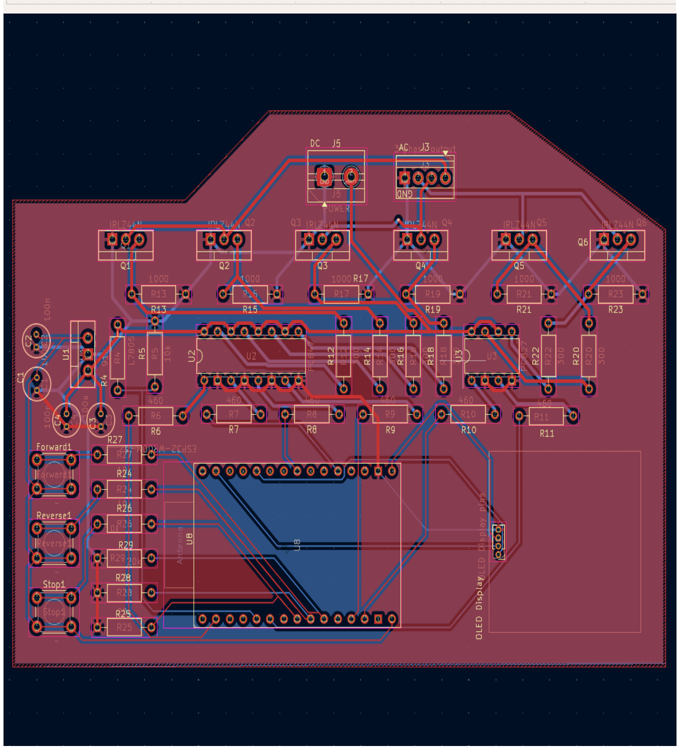
Summary
To expand my knowledge of power electronics, I decided to design and build a three-phase inverter. I chose a three-phase topology to challenge myself with more complex switching and control techniques while gaining experience with a system that can be scaled to applications such as motor drives and renewable energy.





1) System Design & Architecture

- Six MOSFETs form the three-phase inverter bridge.
- Optocouplers electrically isolate the power stage from the ESP32 control circuitry.
- A 5V regulator supplies the low-voltage electronics.
- DC link and decoupling capacitors help stabilize the supply and reduce high-frequency noise.
- A voltage-divider circuit scales the battery voltage for safe measurement by the ESP32 ADC.
- Selected a four-channel optocoupler instead of individual single-channel devices to reduce PCB area, component count, and routing complexity.

2) Embedded Control & MCPWM:

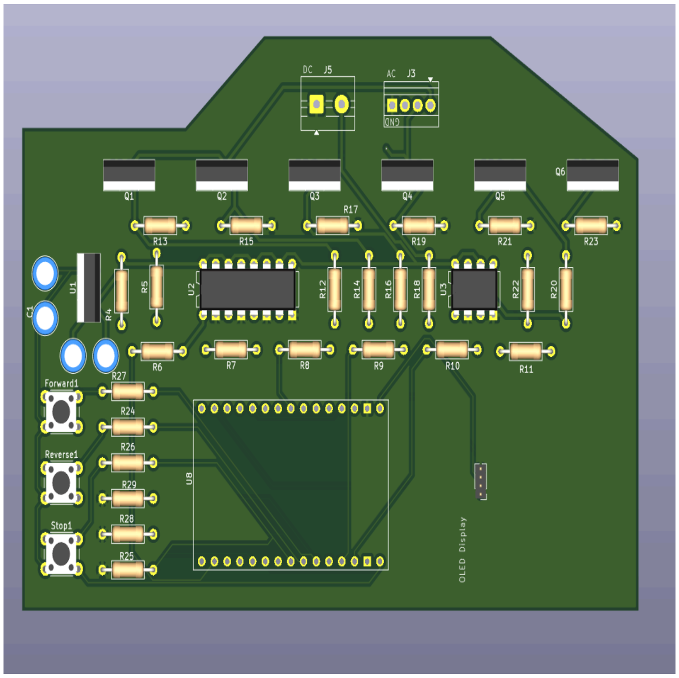
- Implemented dead-time insertion between complementary switching signals to prevent shoot-through.
- Implemented fault-detection logic to safely disable the inverter during fault conditions.
- Investigated SPWM and SVPWM as more advanced modulation techniques.

3) Testing & Debugging

- Used an oscilloscope to verify PWM frequency, duty cycle, complementary signals, dead time, and switching waveforms.
- Used a multimeter to verify regulator output, battery voltage, voltage-divider measurements, and circuit continuity.
- Troubleshot ESP32 GPIO limitations and MCPWM configuration issues.
- Debugged hardware/software integration problems by comparing programmed behavior with measured waveforms.
- Used iterative testing to identify and correct circuit and control issues.

4) Take Away:

- Expanded my understanding of three-phase power conversion and motor-control systems,
- Gained experience making hardware design tradeoffs involving isolation, PCB space, noise, and system reliability.



5) Future improvements:

- Implement SVPWM and closed-loop current control.
- Add current, temperature, and overvoltage/undervoltage sensing.
- Integrate a dedicated gate-driver IC.
- Improve thermal management and PCB creepage/clearance.
- Characterize inverter efficiency, switching losses, and output waveform quality.